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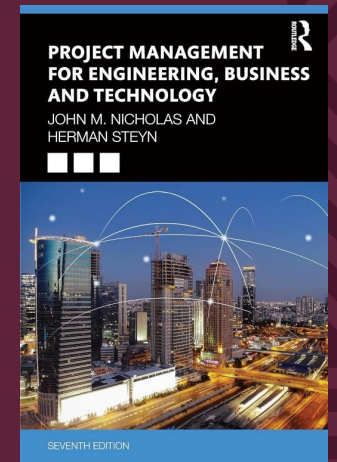
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# CH04: PROJECT AND SYSTEM DEFINITION

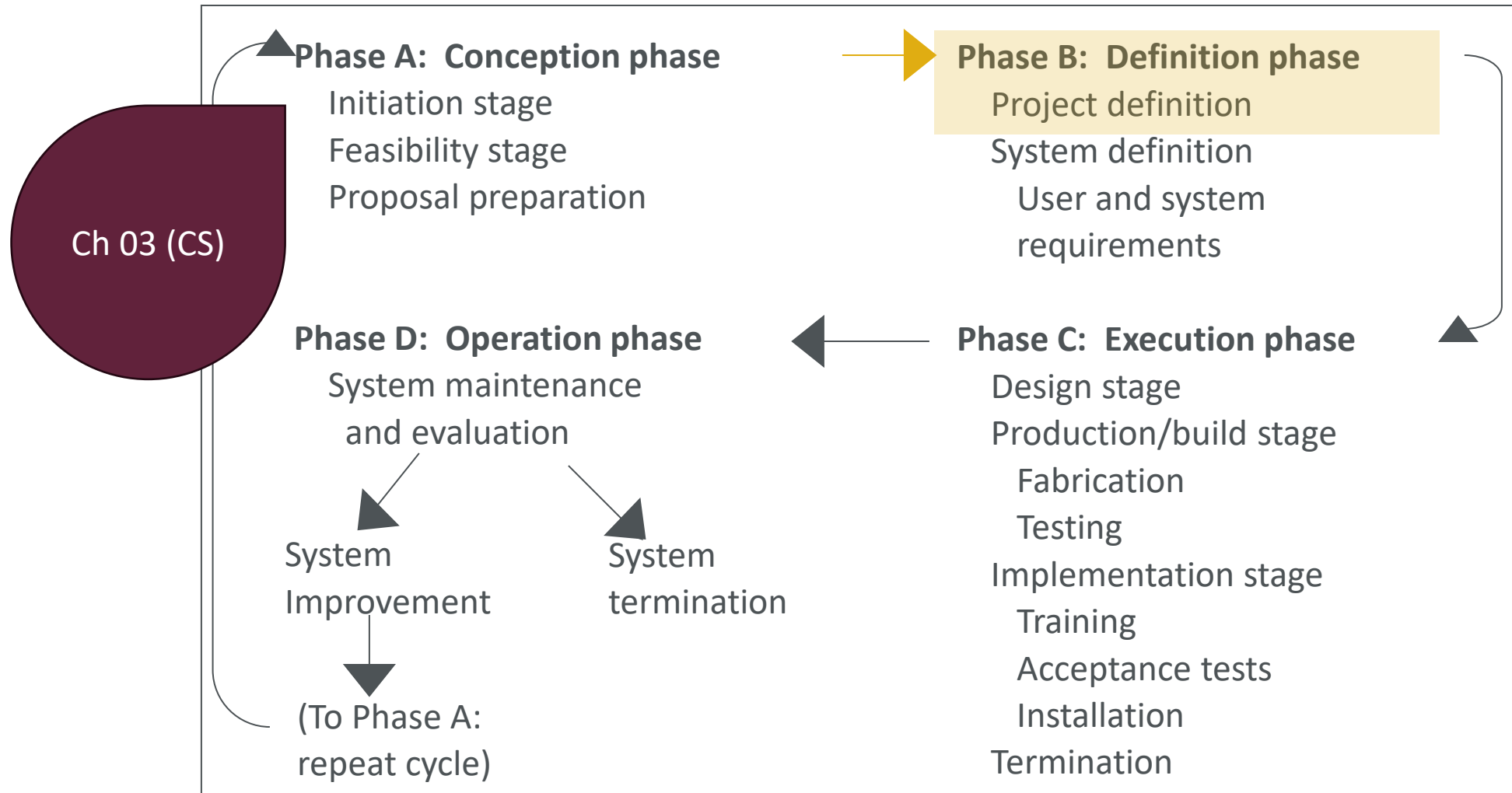
Ms Anthea Venter

Adapted from slides  
prepared by

*John Nicholas and  
Herman Steyn*



# Systems Development Cycle or Project Life Cycle



# Phase B: Definition



- Assume upon entering this stage
  - Project has **been approved** and **funded**
- Also, assume these **exist**
  - An **SOW** in RFP and Proposal documents
  - Initial list of user **requirements**
  - A “rudimentary” **project plan**, as necessary for specifying technical content, time, and price in the proposal
  - **Contract** with a SOW (CSOW)

# Phase B: Definition (cont'd)



- Principle tasks during Phase B  
(not necessarily in this order)
  - **Organize** project team: hold “kickoff”
  - **Clarify** in detail user requirements
  - Prepare detailed **system requirements**
  - Prepare **project** master plan
  - **Review** requirements and plan with customer

# Project vs System Definition



## Project Definition:

- Specify *what the project team must do* to produce the end-item
- Create a project plan to direct the project team

## System Definition:

- Achieve a good understanding of *what the end-item must do to satisfy user requirements*
- Create detailed requirements and specifications for end-item
- Provide these to the project team so they know what they have to produce

# Phase B: Definition Tasks



- In **little** projects, Phase B is short since
  - much definition already happened in proposal preparation
- In **big** projects, Phase B can be lengthy
  - sometimes taking months/years

# Project Kickoff Meeting



- The **first** formal meeting of the project team members and key stakeholders.
- A formal **presentation** with a question-and-answer period at the end.
- The **project manager** plans and runs the meeting.
- Runs 1.5-2 hours
- Purpose is to **announce** the project:
  - communicate what the project is **about**
  - develop common **expectations**
  - generate **enthusiasm** and commitment to project goals and deliverables

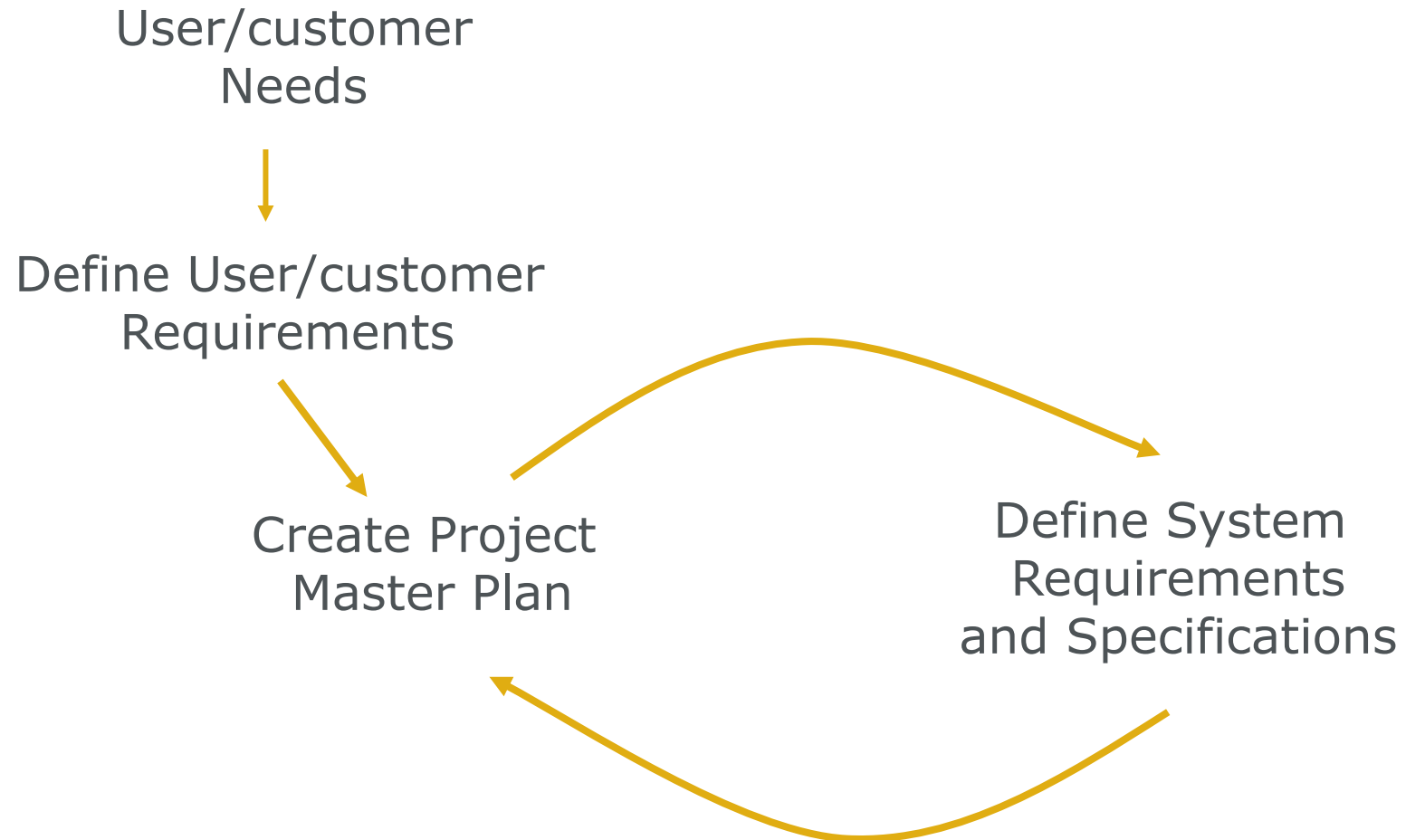
# Project Kickoff Meeting (...)



- Covers
  - who is the project manager
  - project SOW, goals, and deliverables
  - proposed project plan—budget, schedule, main work packages
  - constraints and risks
  - customers and other key stakeholders, their needs and requirements
  - project organization structure and key team members
  - immediate next steps
- Held for **every** project and every major effort associated with the project

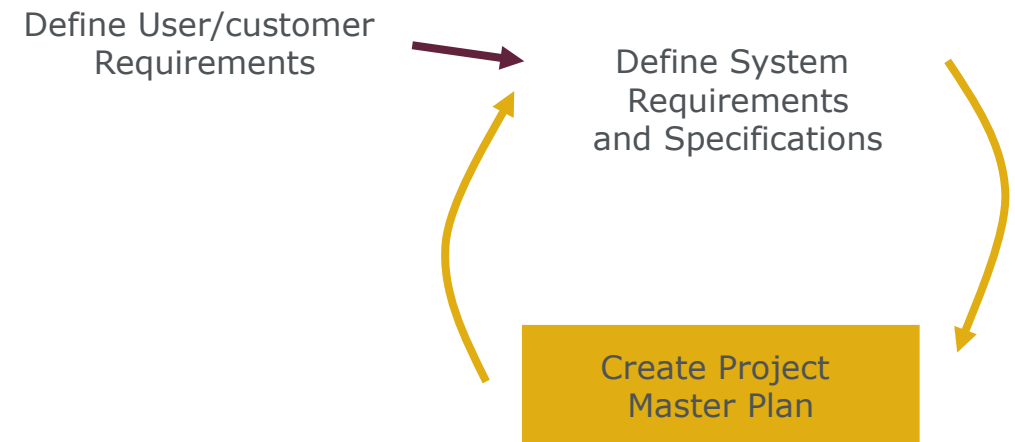


# Phase B: Primary Definition Tasks



# Project Definition

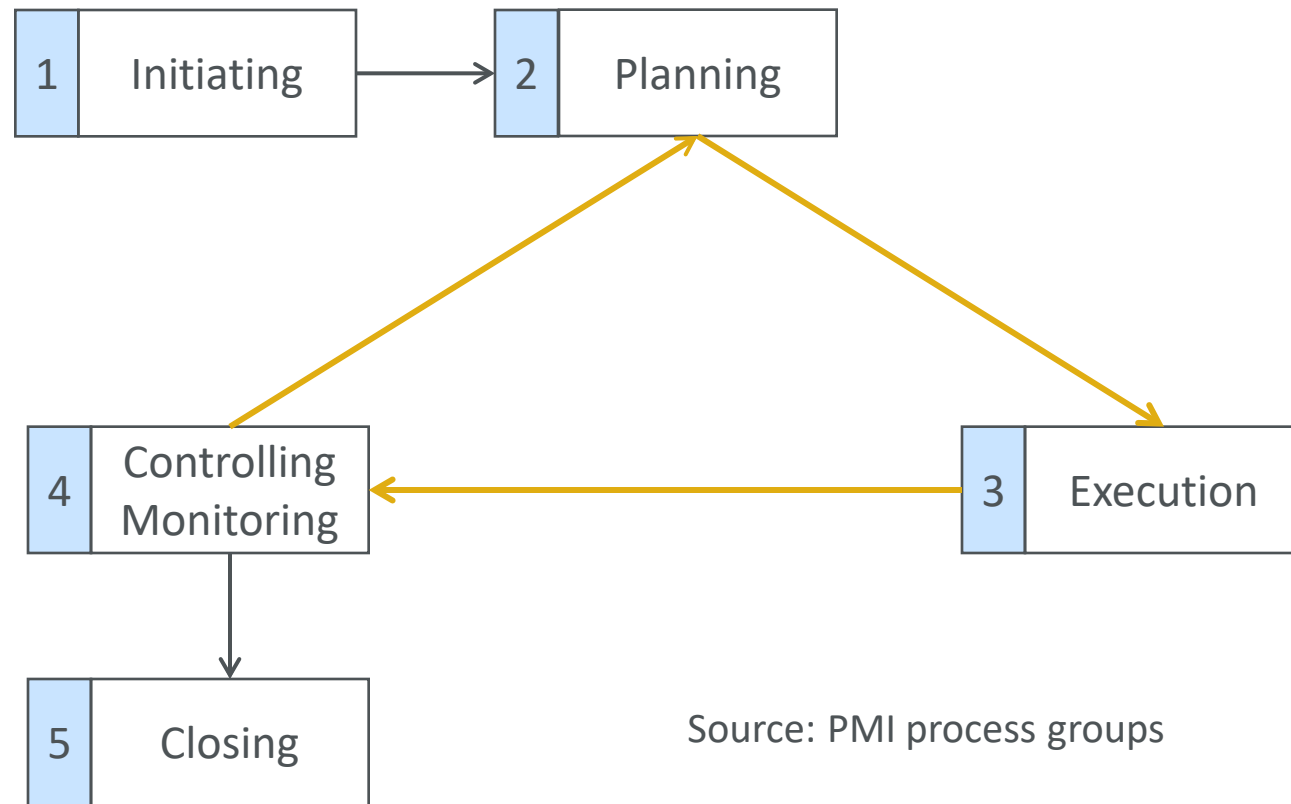
- During Definition, the **project execution plan** and end-item requirements and specifications are defined.
- The system requirements and specification address “what” the end-item of the project must do.
- The project master plan describes “how” project will deliver end-item that meets system requirements and specifications



# Iteration is Key!

- It should be an iterative process
- Details of the specifications are defined; master plan is expanded to reflect details
- As master plan is expanded, project constraints, opportunities, resources are identified, which leads to revisions in specifications

# Project Definition = Project Planning



# Project Definition



- What goes **into** a project plan (the “execution plan”)?  
(**Proposal** addressed these questions, but usually not in much detail)
- Ask:
  - What?
  - How?
  - Who?
  - When?
  - How long?
  - Where?
  - How much?
  - How well?

# Project Master Plan



- Project execution plan addresses these questions to the satisfaction of **project core team** (people who will do work)
- Addresses all matters about project in **sufficient detail** for managers to organize and direct work to meet performance, cost, and time targets and for team to **begin** work
- **Level of detail** in the master plan far exceeds level in the proposal

# Common Elements of Project **Master/Execution Plan**



1. **What?** Scope Statement, Charter, or SOW
2. **What?** Detailed requirements
3. **How?** Detailed work definition (WBS or PBS and work package/work task details)
4. **Who?** Responsibility for work tasks
5. **When?** Detailed schedules with milestones
6. **How much?** Project budget and cost accounts

# Common Elements of Project Plan (cont'd)



7. What if? Risk plan
8. How well, what, how? Performance tracking and control
9. Other elements of the plan, as needed for, e.g.
  - Work review and testing
  - Quality control
  - Documentation implementation
  - Communication/meetings
  - Procurement
  - Contracting and contract administration

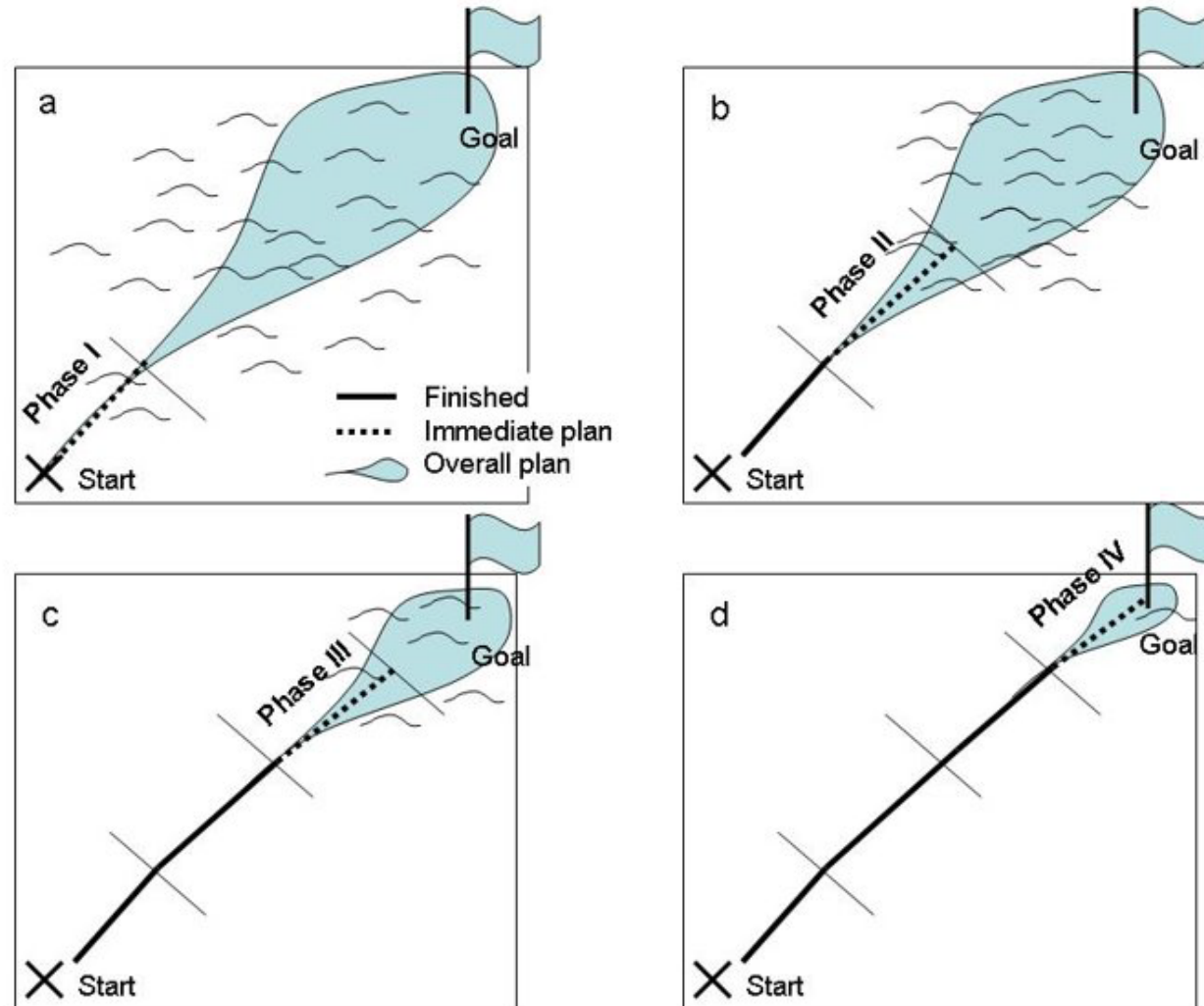


# Phased (Rolling Wave) Project Planning



- At the start of the project, often there are too many unknowns, so the plan must be developed in phases
- The initial plan is somewhat rough though adequate to
  - estimate project resources, time, and cost
  - explain all this to the customer
- As the project progresses,
  - the unknowns decrease
  - details of the plan are filled in
  - a more-detailed plan is created for the next most immediate phase of the project
- As project moves through the successive phases and stages, detailed plans are prepared with more-specific deliverables and schedules.

# Phased Project Planning



# Phased Project Planning



- Sometimes each phase **concludes** with a **milestone**
  - The customer or management **review** the deliverables and project performance
  - **If satisfied**, they approve the deliverables and pay for work done thus far.
- They also **review** the detailed plan for the **next phase**,
  - If satisfied they authorize the next phase.
- Authorization to begin the next phase represents a **commitment** by the customer and management to support the phase
- If the project must be **terminated**, it is terminated at the end of a phase.

# Project Charter



- For *internal* projects, the charter describes the project to stakeholders.
- Sometimes it is used to generate interest in a proposed project.
- Often it is used to announce authorization of an approved project in the organization and establish the project manager's authority to gather and make use of resources.
- It provides a good overview of the project and may include
  - the project objectives and scope
  - stakeholders and their stakes
  - estimated budget and schedule
  - risks
  - assumptions and constraints
  - resources
  - key roles and the people them
- Sometimes it is used as the project plan; more commonly it is somewhat brief

# Project Charter



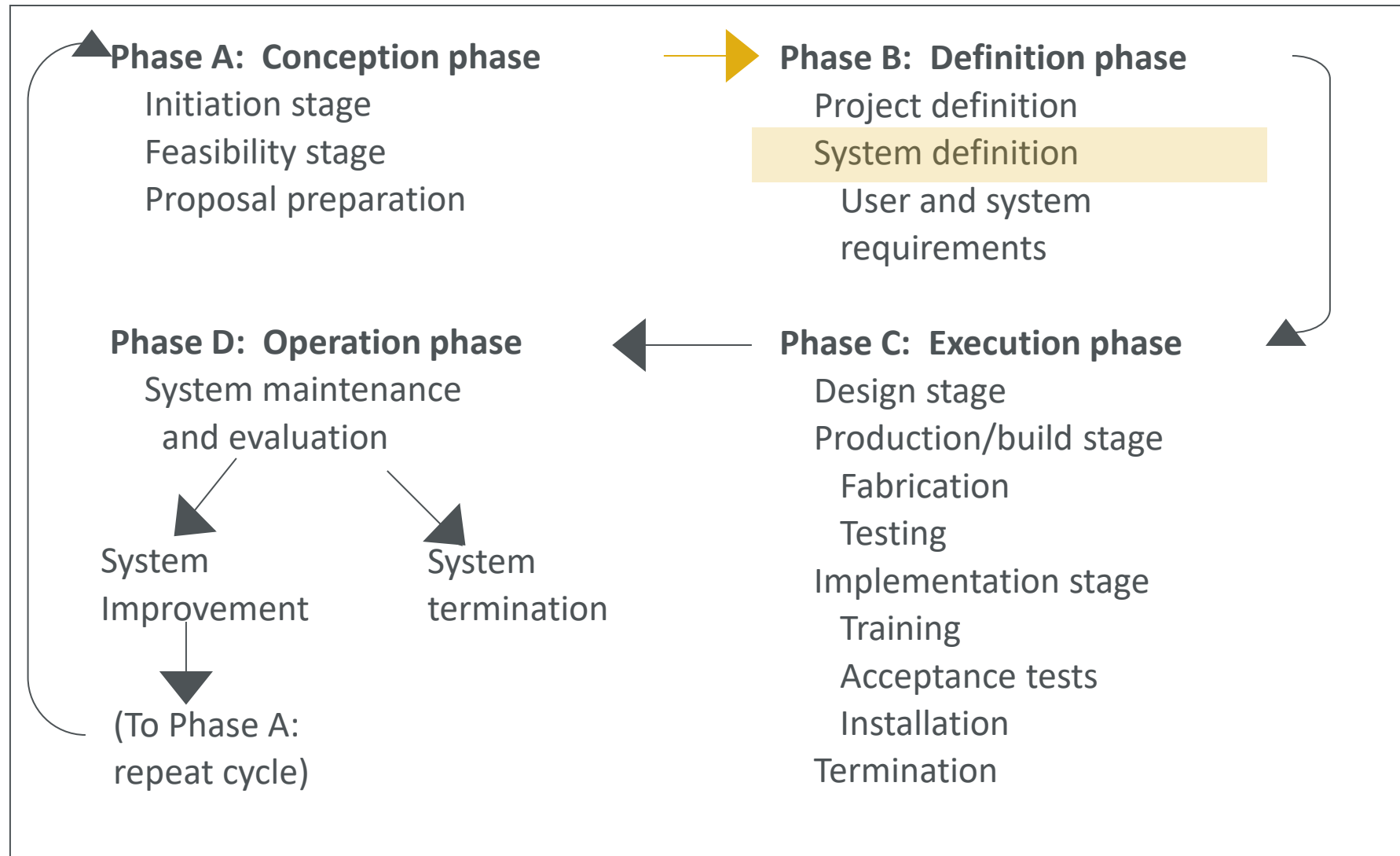
- May include everything in Scope Statement plus
  - risk limits
  - customer needs
  - spending limits
  - key players on project team.
- Issued by senior management to legitimize project
- Gives project manager authority to initiate work and apply resources to project.

# Charter Contents



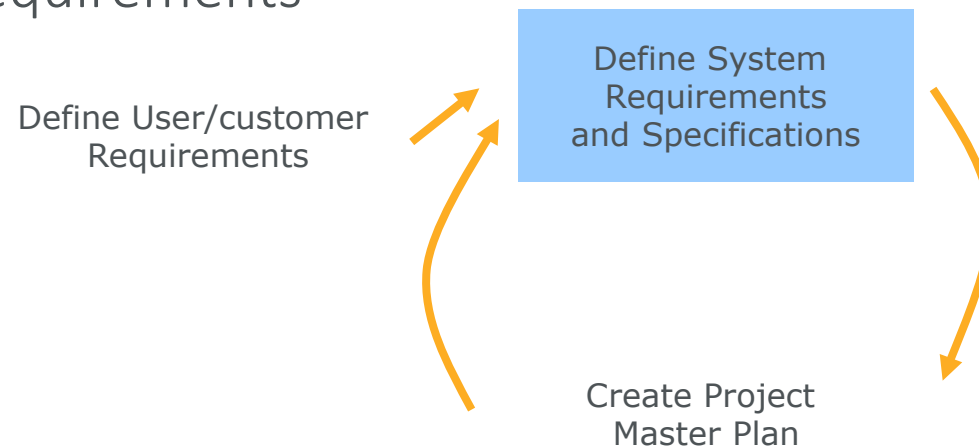
- Background
- Project Objectives
- Scope or SOW
- Deliverables
- Assumptions
- Constraints
- Approach
- Schedule
- Project Team
- Risk
- Management Plan

# Systems Development Cycle or Project Life Cycle



# System Definition

- **System requirements** and specifications elaborate in detail on the technical performance of end-item
- Tell designers and builders what **project end-item** (deliverable) must be and do
- Are a translation of **user** requirements into **technical** requirements
- Users are **ignorant** of most system requirements





# Requirements Definition is Important!



*Project failure often stems from ambiguous or incomplete requirements*

Example:

*“I want this room painted blue”*

# Requirements Definition is Important!



This statement is ambiguous and incomplete.

- Doesn't say what *type* of blue or *how much* of room to paint.
- Should have specified exact color (paint spec. #) and exact part of room (e.g., "only walls")

# Requirements Definition is Important!



- Without clear requirements, contractor
  - Cannot know “what” is wanted
  - Hence, cannot know how to provide it
  - Hence, cannot define the necessary project work (“how” the project must be done)

# Start with User Requirements



- Most customers are somewhat unclear as to what their requirements are.
- Role of PM is to work with customer/user to clearly define requirements. This is a contractor responsibility since the project must fulfill customer's requirements.

# Requirements Definition



- Requirements are
  - the “whats” that the project seeks to provide
  - the basis for project planning
  - the basis for determining project completion
  - define the contractor’s obligation to customer
  - a principle cause of project cost and schedule overruns
- Despite their importance, good requirements are not necessarily easy to create...

# Problems with Requirements Definition



1. **Incorrect Requirements:** Wrong Needs (“Needs” = user’s/customer’s problem)
  - Incorrect Definition of Needs
  - Shifting or Vagueness of Needs
  - Needs of Wrong User
  - Conflicting Needs of Multiple Users
  - Distortions of Needs by Experts

# Problems with Requirements Definition (cont'd)

## 2. Imprecise or Ambiguous Requirements (Subject to Multiple Interpretations)

- Human Language
- Deliberate Imprecision for Flexibility
- **Nebulous** Projects
- User's Lack of Expertise
- Project Planner's Oversight



nebulous

/ˈnebjʊləs/

adjective

1. in the form of a cloud or haze; hazy.

"a giant nebulous glow"

Similar:

indistinct

indefinite

unclear

vague

hazy

cloudy

2. (of a concept) vague or ill-defined.

"nebulous concepts like quality of life"

Similar:

vague

ill-defined

unclear

hazy

uncertain

indefinite

# Problems with Requirements Definition (cont'd)



## 3. Shifting Requirements

- User's Change of Mind
- Insurmountable Obstacles
- New Opportunities
- Seeking Perfection

## 4. Over-Specification of Requirements

- Initiative Discouraged
- Requirements Ignored
- Insufficient Information

## 5. Under-Specification of Requirements

- Chaotic project planning resulting in cost and schedule overruns



# Guidelines for Defining User Requirements



1. State **each** requirement clearly; have both user and project staff **sign-off** on it
2. Assume if a requirement **can be misinterpreted, it will** be misinterpreted
3. Accept that **changes** to project are **inevitable** and things will not go precisely as planned
4. **Include** pictures, graphs, models, and other non-verbal **exhibits** in requirements formulation
5. Carefully **monitor changes** to requirements once project has begun
6. **Educate** both user and project staff about problems associated with specifying requirements

# System Requirements Definition



- System requirements and specifications are a translation of user requirements into technical requirements
- They elaborate in detail on the technical performance of end-item
- They tell designers and builders what project end-item (deliverable) must be and do
- Often, users are ignorant of system requirements (in most cases, that's okay)

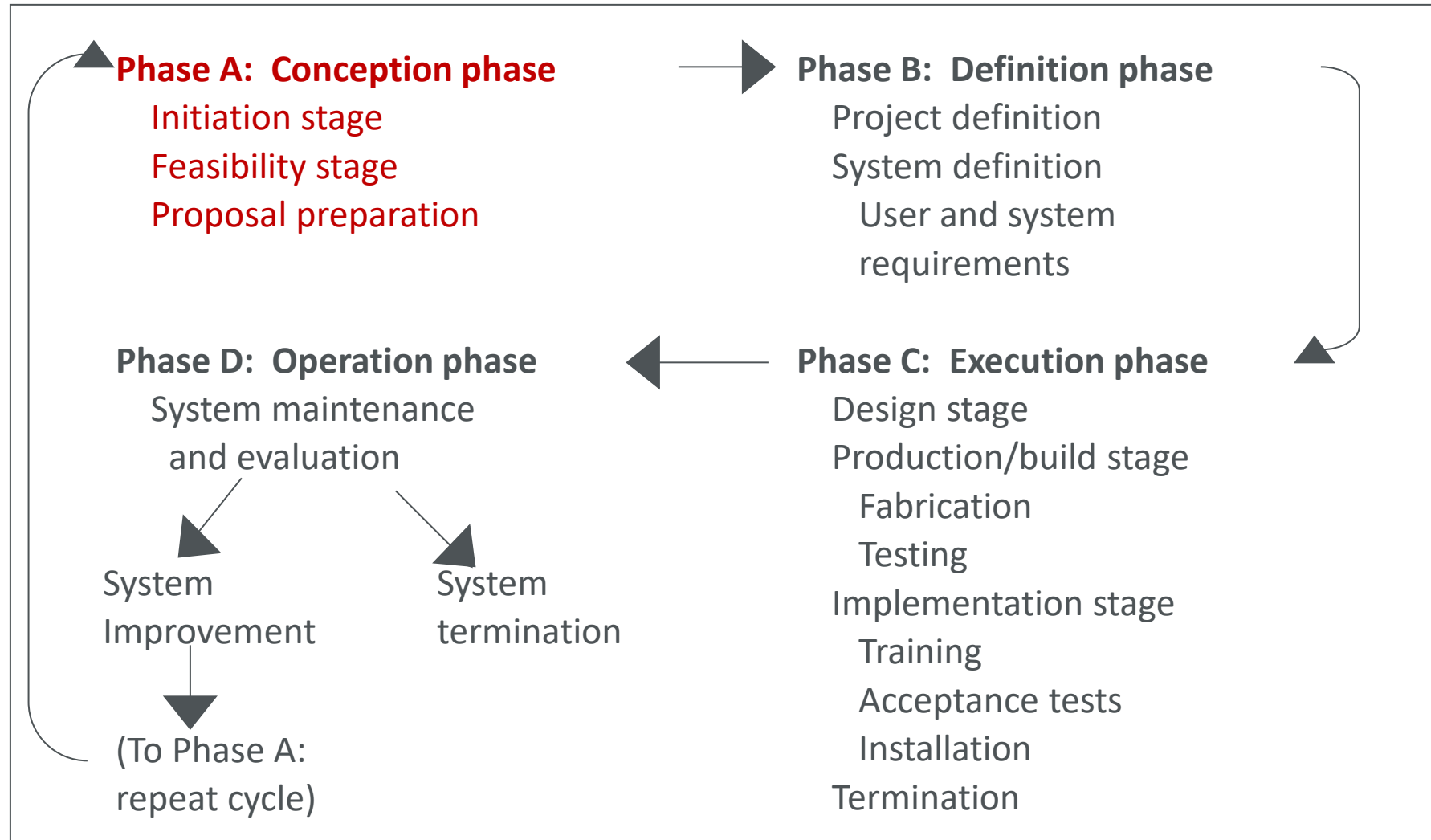
# Requirements Definition, Example



- Customer/user need

Reusable, three-person space vehicle that can be launched twice within two weeks

# Systems Development Cycle or Project Life Cycle

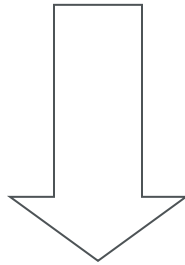


# Requirements Definition



## Customer/user need

- Reusable three-person space vehicle that can be launched twice within two weeks



## Customer/user objectives and constraints

- Win X-Prize (\$10 M)
- Develop technology upon which to build an entire system to carry paying passengers into space.
- Develop vehicle that will be FAA certified.

# Requirements Definition

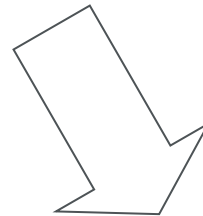
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## Customer/user requirements, examples

- 1 Climb to 100 km.
- 2 Comfortable, enjoyable flight
- 3 Capable of 2-week turnaround, etc.

# Requirements Definition



## Customer/user requirements, examples

- 1 Climb to 100 km.
- 2 Comfortable, enjoyable flight
- 3 Capable of 2-week turnaround etc.
  
- These requirements must be translated into system requirements
- System requirements are the **technical** requirements
- They tell the project team what the end-item system must do: **System Specifications**
  - A **functional requirement** is a kind of system requirement. It specifies the functions the end-item system must perform to meet the user requirements
  - Associated with functional requirements are **performance requirements** that specify the required level of performance
  - **Verification**: How you would measure compliance to the Customer/User requirements

## Customer need

Reusable three-person space vehicle that can be launched twice within two weeks



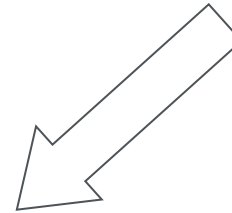
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## Customer requirements, examples

- 1.0 Climb to 100 km.
- 2.0 Comfortable, enjoyable flight
- 3.0 Capable of 2-week turnaround etc.



## System requirements, examples

- functional
- performance
- verification

### 1 Climb to 100 km.

1.1 Function: Engine generates enough **thrust**

**Performance:** 73,500 kN (16,523 lbf).

**Verification:** simulation; mockup tests; ground tests of ignition, ramp up, steady state, shut down



Specifications

### 2 Comfortable, enjoyable flight

2.1 Function: Cabin **temperature** at comfortable level

**Performance:** 75-85 degrees F

**Verification:** ground tests, extreme environment chamber; flight tests

### 3 2-week turnaround

3.1 Function: **Refueling** takes less than 2 weeks

**Performance:** Actual refueling procedure should take 3 days max

**Verification:** simulated refueling procedure; refueling tests, etc.



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# Requirements Priority and Margin



Each requirement should have a specified priority and margin.

## Priority Level

- The relative importance of the requirement
- In case multiple requirements conflict, the priority level determines which can be bent and which not

## Margin

- The amount by which the requirement can vary. For example, “max temperature 85 degrees F ; margin 2 degrees ” says that, if necessary, max temperature can be exceeded by up to 2 degrees.

# Requirements Definition

## Customer need

Reusable three-person space vehicle that can be launched twice within two weeks



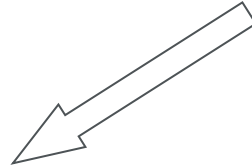
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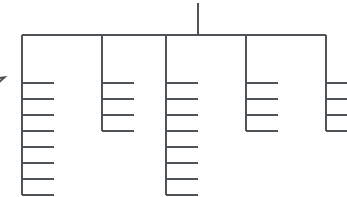
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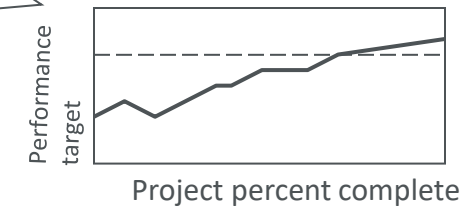
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- 3.1 Function: Refueling takes less than 2 weeks  
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## Requirements Breakdown Structure



## Technical Performance Measurements



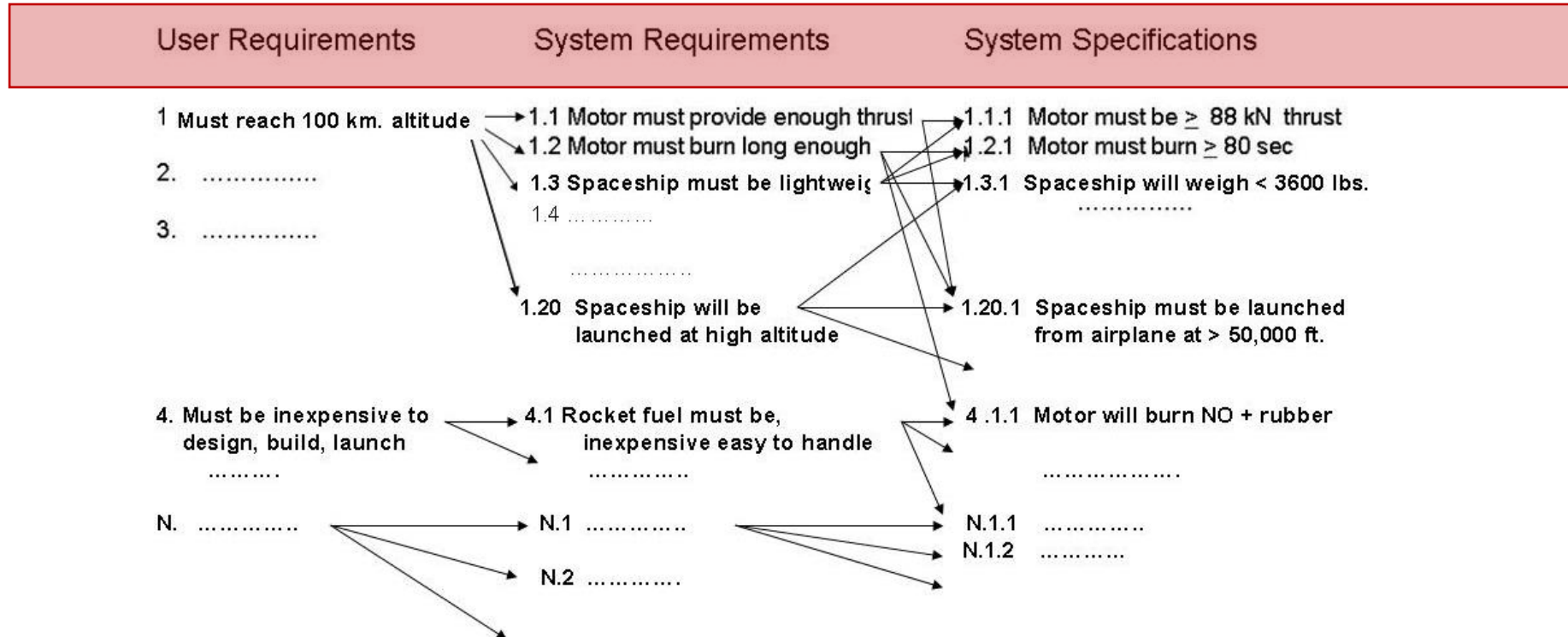
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# System Specifications

- Define in more detail the **system requirements**
  - Example shows system specifications **derived from** system requirements, which are **derived from** user requirements.



# System (Performance) Specifications



- **Guide** actual project work; are written by and for project specialists—systems analysts, programmers, engineers, product and process designers, consultants, etc.
- Address **all areas** of the project—design, fabrication, installation, operation, and maintenance.
- Enable **traceability**
  - Throughout the systems development cycle numerous **changes and tradeoffs** will be made to requirements and specifications
  - **Tracing** the impact of changes in some specifications and requirements to others is called “traceability”
  - **Traceability** involves keeping track of specifications, tying them to physical components, tracing their impacts, and controlling changes so requirements are met and do not conflict
  - Managing traceability is called **configuration management** and **change control**

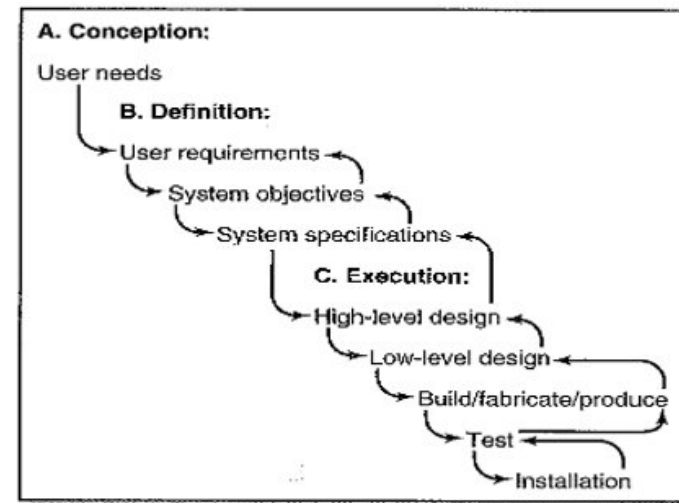
# Alternate Approaches to Definition and Execution



- Traditional (or Waterfall) methodology—follow phases and stages in sequence
  - PMI calls this the Predictive approach
- Agile methodology—execute phases and stages iteratively
  - PMI calls this the Adaptive approach

# Waterfall Development Methodology

- Requirements definition and system design and testing: happen iteratively, particularly when the project end-item is complex.
- Requirements cannot be defined without some amount of prior design work, and design work cannot be completed without some amount of prior fabrication and testing.
- Overall process cascades down, with back-loops and repeated steps. Work that flows like this is called the *waterfall process*.



# Rapid Prototyping



- It can be difficult to conceptualize a system when no system exists like the one to be developed.
- Requiring the customer to specify and sign-off on requirements early in the project is risky.
- Rapid prototyping:
  - A rapid prototype (RP) model is prepared. It is a rudimentary incomplete model that initially is somewhat simple and inexpensive to produce.
  - It represents *key parts* of the system but not the complete system and is somewhat easy to create and modify.
  - The customer experiments with the RP to assess the system's functionality and determine any modifications or additions.
  - After a few iterations of experimenting and modifying requirements, the final requirements and design concept are firmed up.
- RP process ensures that drawings and requirements are finalized only after the customer has accepted them as represented by the RP model.

# Agile Project Management (elaborated in Chapter 14)



**Agile Project Management** is a variant of *iterative life cycle* and *agile management* where deliverables are *submitted in stages*.

- End-item is divided into pieces so that the functionality of the end-item is developed and delivered in increments
- Agile applies to development projects, especially software where the functionality of the end-item system can be achieved piecemeal through incremental development of modules to the system
- Breaks the project into small (modularized) pieces
- Emphasis on delivering the **smallest “workable piece”** of functionality and business value early on and continually improving and adding functionality throughout the life of the project



# Why Agile Project Management?



For certain kinds of projects, agile can

- Reduce development costs
- Improve software reliability
- Decrease time to development
- Reduce the number of errors developers make
- Eliminate the cost of a failed end-item system.
- Pieces of the system that have been developed *will work*, even if other intended pieces of the system are not yet in place.
- The customer receives workable results that can be put into operation, *even if the project is cancelled*.

# Project and System Definition



How do you keep everyone in the project focused on those requirements?

How do you **develop a project plan** that will be able to account for those requirements?

- A: make the system and project definition a **team effort**
  - incorporate the perspectives of **everyone** with a stake in the project
    - customers, suppliers, functional areas such as engineering, marketing, manufacturing, customer service, and purchasing, and users and operators.
- The more these **individuals** and **groups** have a hand in defining requirements and the plan, better the system requirements will account for their needs throughout the systems life cycle

# Project Communication and Communication Plan



- Everything in project management involves communication!
- Communication plan addresses all forms of communication—formal, informal, verbal and written
- Includes tentative schedule for formal design and management review meetings, milestone meetings, etc.
  - Describes meeting formats, itineraries, preparations, attendance, and leader.
- Gives points of contact: customer, contractor, vendors, subcontractors, supporters, others, and kinds of communication needed for each (next slide)



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| Role/type<br><br>Meeting/<br>report | Status<br>meeting<br>(frequency) | Status<br>meeting minutes<br>(frequency) | Business<br>feasibility | Information<br>request | Technical<br>feasibility | Business<br>brief | Project plan<br>(frequency) | Problems<br>and issues<br>(frequency) | Business<br>study | Use case<br>analysis | System<br>architecture | Detailed<br>technical<br>design | Other |
|-------------------------------------|----------------------------------|------------------------------------------|-------------------------|------------------------|--------------------------|-------------------|-----------------------------|---------------------------------------|-------------------|----------------------|------------------------|---------------------------------|-------|
| Client                              | X                                | X                                        | X                       |                        |                          | X                 | X                           | X                                     | X                 | X                    |                        |                                 |       |
| Relationship<br>manager             |                                  | X                                        | X                       | X                      | X                        | X                 | X                           | X                                     | X                 | X                    | X                      |                                 |       |
| Business analyst                    | X                                | X                                        | X                       |                        |                          | X                 | X                           | X                                     | X                 | X                    | X                      |                                 |       |
| Project manager                     | X                                | X                                        | X                       | X                      | X                        | X                 | X                           | X                                     | X                 | X                    | X                      |                                 | X     |
| Client project<br>team              | X                                | X                                        | X                       |                        |                          | X                 | X                           | X                                     | X                 | X                    |                        |                                 |       |
| IT project team                     | X                                | X                                        |                         |                        |                          |                   | X                           | X                                     | X                 | X                    | X                      |                                 | X     |
| Client director                     |                                  | X                                        | X                       |                        |                          | X                 |                             | X                                     | X                 |                      |                        |                                 | X     |
| IT director                         |                                  | X                                        | X                       | X                      | X                        | X                 | X                           | X                                     | X                 | X                    | X                      |                                 |       |
| Project sponsor                     |                                  | X                                        | X                       |                        |                          | X                 |                             |                                       | X                 |                      |                        |                                 |       |
| IT VP                               |                                  | X                                        | X                       |                        |                          |                   |                             |                                       | X                 |                      |                        |                                 |       |
| Architect                           | X                                | X                                        | X                       | X                      | X                        | X                 | X                           | X                                     | X                 | X                    | X                      |                                 | X     |
| Security/audit                      | X                                | X                                        | X                       |                        | X                        |                   |                             | X                                     | X                 | X                    | X                      |                                 |       |
| Internet<br>operations              | X                                | X                                        | X                       |                        | X                        |                   | X                           | X                                     | X                 | X                    | X                      |                                 | X     |
| Intranet<br>operations              | X                                | X                                        | X                       |                        | X                        |                   | X                           | X                                     | X                 |                      |                        |                                 |       |
| Legal/corporations<br>communication |                                  | X                                        | X                       |                        | X                        |                   |                             | X                                     | X                 |                      |                        |                                 |       |
| Other                               |                                  |                                          |                         |                        |                          |                   |                             |                                       |                   |                      |                        |                                 |       |

Sample communication plan.



Thank you  
Enkosi  
Dankie

